

Milan Housing Project Report

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English translation prepared on 27 September 2026. Original figure artwork is retained; English keys accompany the Italian labels.

1 Data cleaning, missing-value imputation and predictor transformations

The `training` and `test` sets contained some missing or inconsistent values. Transformations were also considered appropriate for several predictors. Each dataset field was treated as follows:

- Missing values of `bathrooms_number` were imputed using cumulative logistic regression, with `square_meters` as the predictor.
- `rooms_number` had no missing values. The value 5+ was converted to 6.
- Some `square_meters` values were incompatible with `rooms_number`. They were replaced by predictions from a linear model with `rooms_number` as the predictor. The variable `sqm_1`, equal to the reciprocal of the area in square metres, was also created.
- A new response variable, `sqm_price`, was created from `selling_price`. It is the logarithm of the price per square metre and is used to fit the final model and calculate predictions.
- Missing values of `year_of_construction` were imputed using the mean of the other values. An indicator, `new`, was also created: it equals 1 if the property was built after 2010.
- Missing values of `total_floors_in_building` were imputed using a linear model with `year_of_construction` as the predictor.
- `car_parking` was split into `shared_parking` and `garage_box`, which represent parking spaces in garages and communal areas, respectively.
- `availability` was converted to an indicator equal to 0 if the property is not immediately available and 1 if it is.
- `floor` was converted to a numeric variable. The values `ground`, `mezzanine` and `basement` were set to zero, and corresponding indicators were created. The variable `ratio`, the ratio of floor to `total_floors_in_building`, was also created.
- `energy_efficiency_class` was converted to numeric values. Missing values were predicted using cumulative logistic regression.
- An indicator was created for each feature in `other_features`, including an NA indicator for properties with no reported features.
- `lift` was converted to an indicator equal to 0 if the property has a lift and 1 if it does not. Missing values were predicted using logistic regression. The variable `floor_lift`, the product of `floor` and `lift`, was also created to identify properties on higher floors without a lift.

- `condominium_fees` was converted to numeric values. It was divided by `square_meters`, and missing values were imputed using the mean.
- Missing values of `heating centralized` and `conditions` were imputed using the mode.
- The values of `zone` were geolocated using the Google Maps API and imported into the geographical software QGIS. Geolocation errors were corrected. As shown in Figure 1, a legend was added to show the number of observations for each zone in the `training` set. Zones with fewer than 3 observations in `training` were merged with neighbouring zones.

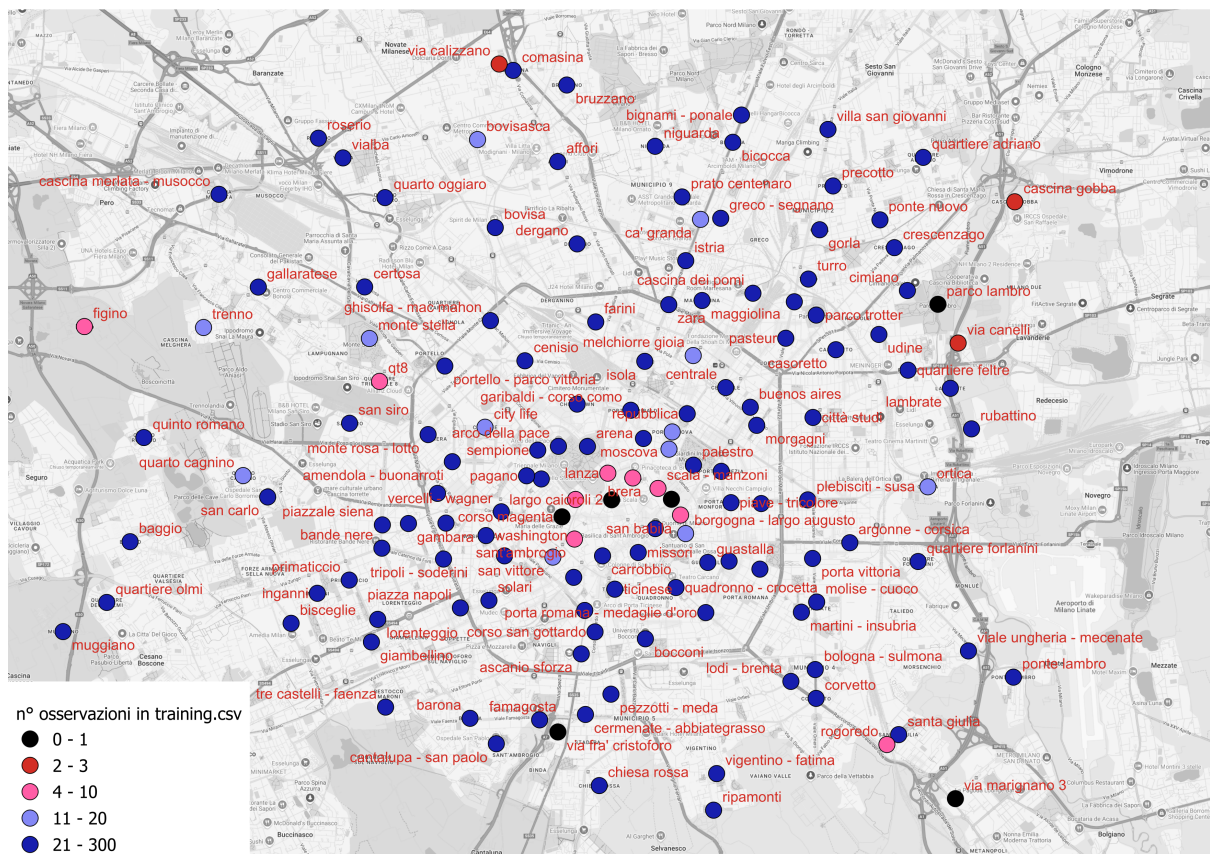


Figure 1: Map of Milan showing the zones present in the datasets. The legend gives the number of **training** observations in each zone.

English key. “n° osservazioni in training.csv” means “number of observations in training.csv”. The categories are 0–1, 2–3, 4–10, 11–20 and 21–300. Neighbourhood and place names are retained as proper names.

2 The model

Because the loss to be minimised was absolute error (MAE), a quantile regression model was chosen. In this project, implementing dimension-reduction or shrinkage techniques such as lasso, group lasso, ridge and PCA with quantile regression proved difficult. Trials of these techniques on the dataset did not yield satisfactory results. Quantile regression was therefore preferred despite its greater computational complexity.

2.1 Attempt to reduce the dimensionality of the zones

An attempt was made to merge zones using k-means on their coordinates. The results were assessed on a validation test subset drawn randomly from **training**, while dividing the observations from each zone proportionally between the training and validation test subsets. A series of regressions was fitted using all predictors, with the zones merged into progressively fewer groups, in a procedure similar to backward regression.

As Figure 2 shows, the error increases almost monotonically as the number of zones decreases. All zones were therefore retained in the subsequent steps.

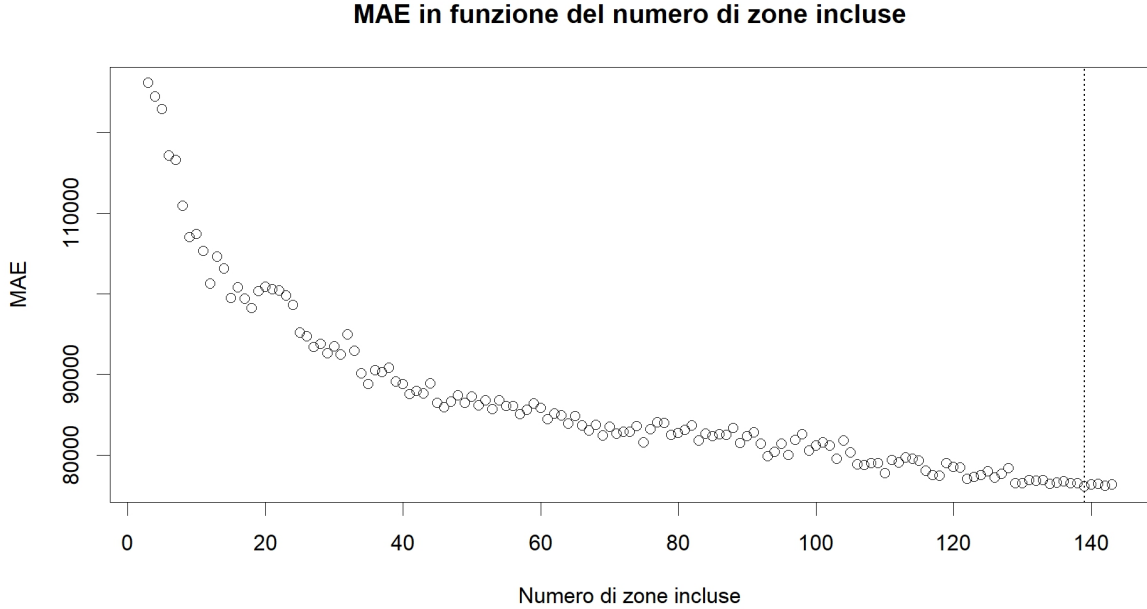


Figure 2: Mean absolute error on the test set as a function of the number of distinct zones included. The number of zones was reduced by merging them using k-means on their geographical coordinates. The vertical line marks the point with the lowest MAE.

English key. “MAE in funzione del numero di zone incluse” means “MAE as a function of the number of zones included”. “Numero di zone incluse” means “Number of zones included”. MAE means mean absolute error.

2.2 Final predictor selection

Backward regression was tried as a way to reduce dimensionality, removing the predictor with the highest p-value at each step. However, the results varied substantially with the choice of training and test sets. Fifty different models were therefore selected using backward regression, and each was evaluated on 50 different training and test splits (Figure 3). The model selected for the final predictions had the lowest MAE and an acceptable variance.

A limitation of this procedure is that the data used for the final evaluation had already been used to select the 50 candidate models. The model's MAE estimate is also biased because that same estimate was used to choose the model.

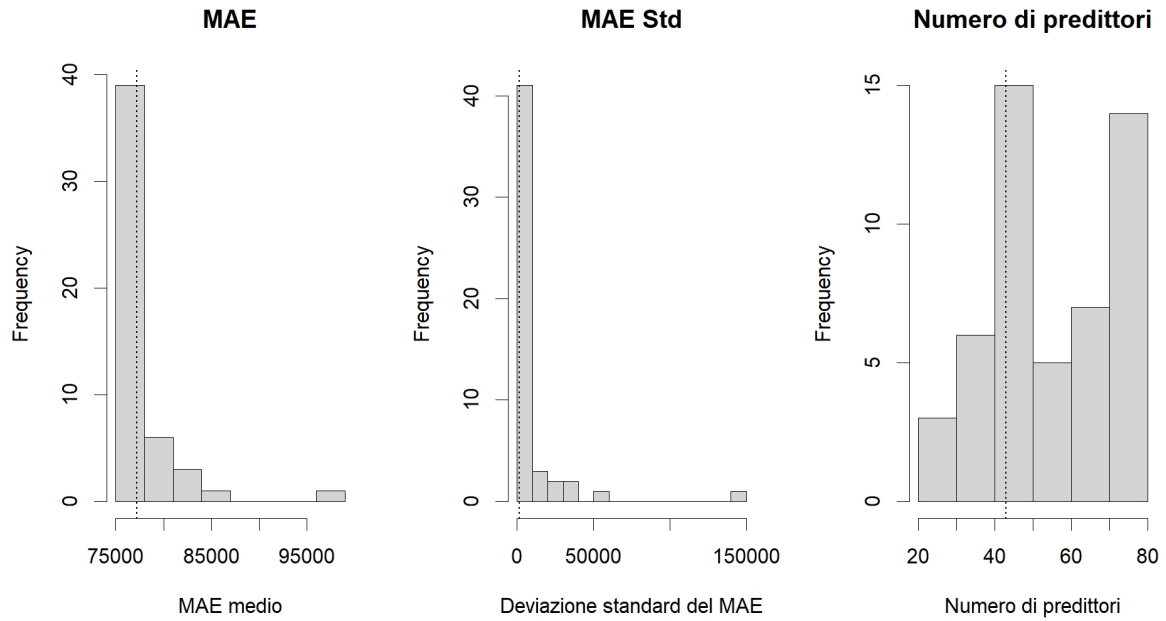


Figure 3: Distributions of the number of predictors, MAE and standard deviation of MAE for 50 models obtained by backward regression on 50 training and test splits and evaluated on another 50 training and test splits. The vertical lines indicate the values for the selected final model. The zone indicators are counted as one predictor, as are the indicators for **conditions**. **English key.** “MAE medio” means “Mean MAE”. “MAE Std” and “Deviazione standard del MAE” mean “Standard deviation of MAE”. “Numero di predittori” means “Number of predictors”. “Frequency” is already in English.